Housing First or Treatment First?

A stylised model, before the quantitative one

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Abstract

This note builds the smallest model in which the paper's two central questions can be asked, and answers them. Who uses drugs, and why are users poor? And what does housing assistance do to drug use? The model has three periods, one asset, and households that differ in permanent income and in nothing else. It is deliberately qualitative: it establishes directions and the relative size of two opposing forces, and every magnitude in the paper comes from the full model that follows.

1 What this model is for, and what it is not for

The full model has six state variables and roughly forty parameters. That is what it takes to answer the welfare and fiscal questions the paper is about, but it is far too much machinery for a reader to see which piece is doing the work. This note strips the environment to the smallest one in which two questions are still well posed.

1. **Who uses.** Drug use is concentrated among low income households. What feature of the environment produces that, and how strong does it have to be?
2. **What housing assistance does.** A voucher pulls drug use in two directions at once. Which force is larger?

It is not for anything else. There is no welfare analysis here, no government budget, no general equilibrium, and no validation section. The calibration is rough on purpose: it puts the model in the right region and no further. Several moments the full model matches are unreachable here in principle, and section 6 says which and why.

2 The environment

Three periods. A household has permanent income y, fixed, distributed lognormally, and differs from other households in nothing else.

Period 1. The household is employed and housed. It chooses non drug consumption, drug use $v \geq 0$, and saving, with flow utility

$$\log c \;+\; \gamma \log h \;+\; \theta\big[\log(b+v) - \log b\big].$$

Using drugs risks a substance use disorder next period, with probability $1 - e^{-\lambda v}$.

Period 2. The disorder d is realised. The household is unemployed with probability u_0, or the higher u_1 if $d = 1$; unemployment pays a replacement rate and a consumption floor catches the bottom. Earnings carry a further penalty ξ if $d = 1$. The household chooses housing and saving. Recovery is ρ when housed and $\rho\,\rho_H$ on the street.

Period 3. Terminal, with the same risks.

Housing. A large unit, a small unit, or the street. The street is free but supplies only h_{hl} of housing services, raises mortality, and cuts the recovery probability. A voucher arrives with probability π_V and pays part of the rent.

Two rental sizes rather than one, for a specific reason. With a single size, a household whose cash falls below the rent cannot be housed at all, whatever the street is like, and the model puts the entire bottom of the income distribution on the street. With a small unit available they downsize instead, which is what happens in the data and what the full model's housing grid is for.

The two policies. Under Housing First the voucher arrives whatever the household's disorder status. Under Treatment First it arrives only if the household is clean. That single difference is the entire policy contrast, and it is what makes the decomposition in section 5 possible.

No preference heterogeneity. Households differ in permanent income and in nothing else, so every household at a given income makes the same choice, and the drug use decision is a *threshold in income* rather than a probability. The sharpness is deliberate. It also keeps intact the paper's claim to generate the income gradient of drug use without ex ante preference heterogeneity: there is none here to do the work.

3 What is calibrated, and what is set

Four parameters are bracketed on four targets. Everything else is set, with a source.

parameter	target	data	model
b, the pull of the first unit	drug participation rate	5.13%	5.12%
p_r, rent per unit of housing	housing spending / income	21.2%	21.3%
h_{hl}, housing on the street	homelessness rate	0.399%	0.25%
λ, the addiction hazard	users who acquire the disorder	33.2%	33.2%

Set externally: the discount factor and interest rate, survival, the housing weight $\gamma = 0.25$, the two rental sizes, the unemployment rate given the disorder at the measured 16.25 per cent, the replacement rate, the consumption floor, recovery and its multiplier on the street, both mortality multipliers, the wage penalty, and the voucher parameters.

One number is worth reporting because nothing was done to obtain it. The share of the homeless population carrying a substance use disorder is not targeted, and the model produces 27.6 per cent against 21.7 per cent in the data. It is the joint prediction of the unemployment channel, the wage penalty and the housing threshold operating together.

The calibration brackets each parameter on its own target in sequence rather than fitting jointly. With no preference heterogeneity the aggregate is a cliff in b: participation is one below a narrow band of values and zero above it. A gradient method steps out of the band, finds a flat surface, and walks to a corner, which is what happened repeatedly before the bracketing was put in.

4 Who uses, and what it takes to make them poor

Start from the force that has to be overcome. Drugs are a normal good bought at a price. A household with twice the income gives up less utility per dollar spent on them, so with *no consequences at all* for acquiring a disorder, drug use concentrates at the top: the average user earns 3.5 times the population average. That is the opposite of the data, where users earn about 95 per cent of the average and households with a disorder earn 69 per cent.

Something has to flip it, and the model has two candidates.

mortality	the disorder kills you, directly, and by putting you on the street, which also kills you and makes recovery harder
resource	it costs you a job, a wage, and the unit you can afford to live in

Each channel is scaled from nothing to its calibrated value, and at every point b is rebracketed so that participation stays at 5.13 per cent. The level of drug use is therefore identical all along both curves, and the only thing moving is *which* households use.

	mortality channel		resource channel	
strength	b	users / average	b	users / average
0%	0.2558	3.47	0.2558	3.47
17%	0.0207	0.62	0.0597	3.37
33%	0.0182	0.26	0.0275	3.37
50%	0.0151	0.21	0.0157	1.69
67%	0.0164	0.19	0.0115	0.33
100%	0.0117	0.19	0.0109	0.19

Three things to read off it.

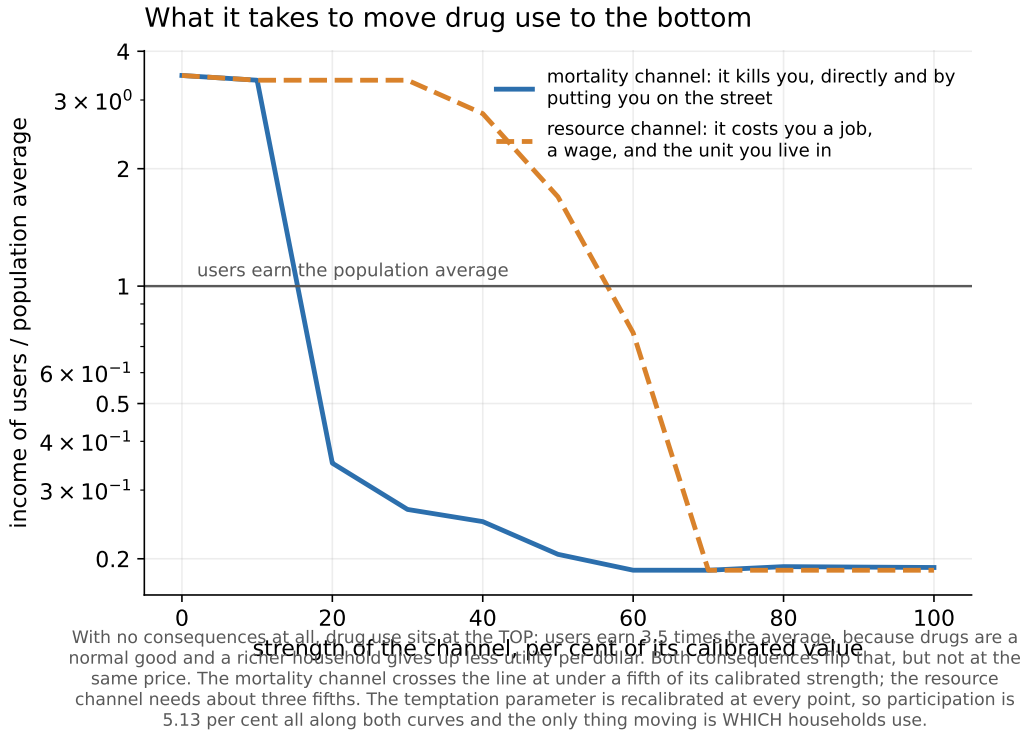


Figure 1: Both consequences move drug use from the top of the income distribution to the bottom, but not at the same price.

The top row is the case with both penalties gone. Both columns start there, because scaling either channel from zero means every consequence of the disorder is switched off. Users earn 3.5 times the average. The model has to be *forced* to put drug use at the bottom; it does not go there on its own.

The recalibration is doing real work, and the b columns show it. With no consequences, a household needs very little temptation to use, and $b = 0.26$ delivers 5 per cent participation. Once the consequences are switched on, the same 5 per cent requires b around 0.011, more than twenty times more temptation. Without rebracketing b at every point, the curves would be comparing economies with wildly different amounts of drug use and the exercise would say nothing about who uses.

Both channels flip the gradient, but mortality does it far cheaper. The mortality channel crosses income neutrality at under a fifth of its calibrated strength. The resource channel needs about three fifths. That is the sense in which the paper's mechanism is the mortality channel: not that it is the only one that can produce the fact, but that it produces it at a fraction of the magnitude.

Why the resource channel works at all, which is not obvious. It is tempting to think a penalty proportional to income cannot flip the gradient, because it scales with the thing it

is supposed to offset. Under log utility that reasoning fails. A proportional wage penalty costs $\log(1-\xi)$ utils whatever the household earns, so it is a *constant* utility loss, and the consumption a household would give up to avoid a constant utility loss rises with its income. It therefore deters the rich more than the poor, exactly as a mortality penalty does. The useful distinction is not mortality against resources; it is constant in utils against proportional in consumption, and under log utility almost nothing is proportional in consumption.

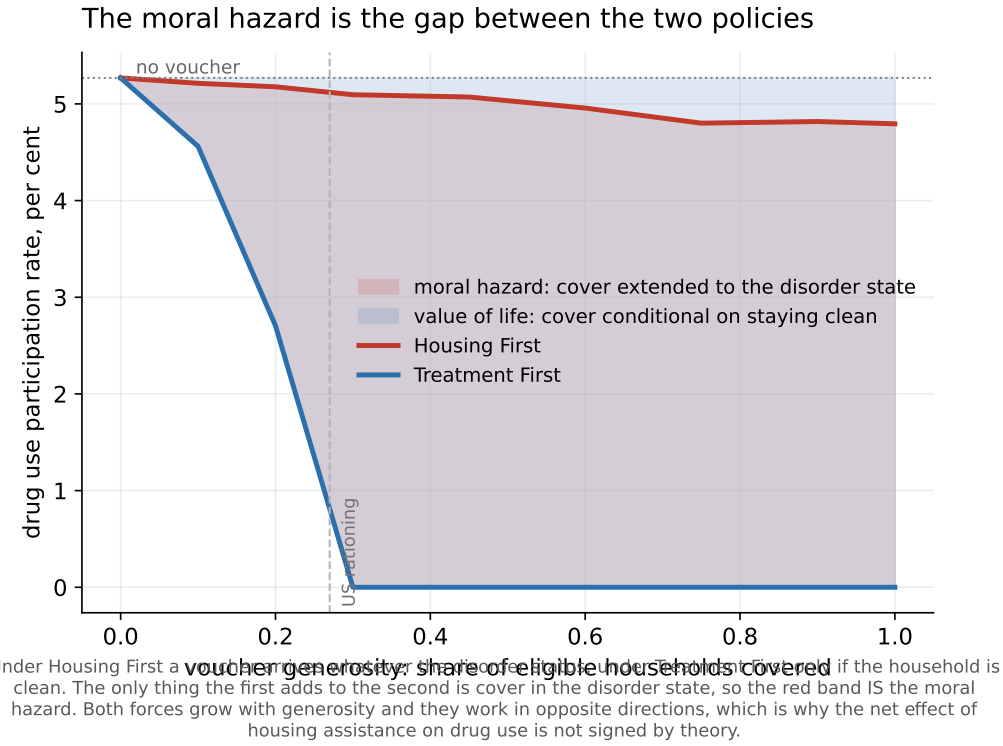
5 What housing assistance does to drug use

A voucher does two things to a household deciding whether to use.

It raises the value of staying clean, because a housed future is worth more than a homeless one, and drug use puts that future at risk. Call this the value of life effect. It pushes use *down*.

It also cushions the downside of acquiring the disorder, because the household that acquires it is housed anyway. Call this the moral hazard. It pushes use *up*.

The two policies separate them exactly. Under Treatment First the voucher is withdrawn if the household acquires the disorder, so only the first force operates. Under Housing First the voucher arrives regardless, so both do. The only thing Housing First adds to Treatment First is cover in the disorder state, which means the gap between them *is* the moral hazard.



Under Housing First a voucher arrives whether the household is using or not. Under Treatment First it arrives only if the household is clean. The only thing the first adds to the second is cover in the disorder state, so the red band IS the moral hazard. Both forces grow with generosity and they work in opposite directions, which is why the net effect of housing assistance on drug use is not signed by theory.

Figure 2: The two forces, as the voucher becomes more generous. The gap between the policies is the moral hazard.

At universal provision, in percentage points of the participation rate:

	participation	against no voucher
no voucher	5.27%	
universal Treatment First	0.00%	value of life −5.27 pp
universal Housing First	4.79%	net −0.47 pp
moral hazard, Housing First against Treatment First		+4.79 pp

Both forces are first order and they nearly cancel. That is the central result of this section, and it is the reason the paper needs a quantitative model at all. Theory does not sign the effect of housing assistance on drug use. Anyone who asserts that housing the homeless will increase drug use, and anyone who asserts it will reduce it, is making a quantitative claim while sounding like they are making a theoretical one.

The value of life effect wins, narrowly. Unconditional Housing First lowers drug use slightly on net, by half a percentage point. This is worth stating plainly because it cuts against the intuition that unconditional assistance must raise use through moral hazard. The moral hazard is real and it is large. It simply loses to the other force.

Conditionality keeps the winning force and removes the losing one. That is the whole argument for Treatment First in one sentence, and it explains something about the previous version of this paper. Its weakest experiment imposed conditionality while holding the number of vouchers at the current rationed level. Both forces scale with generosity, so at the left edge of the figure there is very little for conditionality to work with, while the cost of excluding people from a scarce benefit is immediate. Conditionality is worth more when the benefit being conditioned on is guaranteed.

6 What this model cannot do

Stated plainly, because these are not accidents of the calibration.

The price elasticity of drug participation. With every household at a given income identical, a change in the price of drugs moves the whole threshold and takes a large mass of households with it. The elasticity is pinned by the density of the income distribution at the threshold rather than by any parameter, and it comes out between three and four against roughly 0.45 in the data. Matching it requires dispersion in tastes, which is the thing being deliberately excluded.

Drug spending as a share of income. With one taste, optimal use solves $\theta/(b+v)$ equal to marginal cost exactly, so the level follows once b is set and cannot be targeted separately.

The magnitude of the income gradient. Users earn 19 per cent of the average here against 95 per cent in the data. The direction is right and the comparison across channels is informative; the level is not.

Anything requiring aggregation. No government budget, no market clearing, no welfare. A drug supply curve was written into an earlier version of this model and then removed, because switching market clearing off changed neither of the two results above to two decimal places. It belongs in the full model, where the fiscal and general equilibrium consequences it interacts with actually exist.

7 What the full model adds

Six state variables in place of one, so heterogeneity comes from the environment rather than from assumed taste differences, and the moments this note cannot reach become reachable. A government budget constraint, so policies have to be paid for. An equilibrium drug price. Serious mental illness. An endogenous treatment decision, so that a policy named Treatment First contains treatment. And welfare, which is the object the policy question is actually about.

Two things in this note carry forward as questions rather than answers.

The first is which channel dominates. Section 4 shows that both flip the gradient and that mortality does it at a fraction of the strength. It does not show that mortality dominates at the calibrated parameters, and that is a quantitative question the full model's channel decomposition has to answer rather than assume.

The second is whether the near cancellation in section 5 survives. Here the value of life effect wins by half a percentage point out of five, which is close enough that the ranking could plausibly reverse in a richer environment with treatment, mental illness, and a budget constraint. That the two forces are of comparable size is the robust finding. Which one wins is not.